Neapolitan Artistic Heritage Multimodal Dialogue System

A Graph-Grounded Conversational Agent Integrating DBpedia SPARQL Semantic Harvesting, Dual Graph Persistence, Real-Time Contradiction Reasoning, and Phonetically Robust Speech Synthesis

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Abstract

Deploying conversational artificial intelligence within digital cultural heritage informatics demands deterministic factual reliability, strict geographical domain fidelity, and resilient acoustic parsing of natural human speech. Standard generative large language models frequently suffer from unconstrained hallucinations, anachronistic attributions, and geographical hallucinations when queried about localized art historical collections. In this research monograph, we present the design, semantic formalization, and empirical evaluation of an end-to-end multimodal dialogue architecture dedicated to the Neapolitan artistic heritage. Centered on the foundational Baroque epoch in Naples (1606–1635), the system models the masterworks of Michelangelo Merisi da Caravaggio and his foremost Neapolitan adherent, Giovanni Battista Caracciolo (Battistello), preserved across five urban institutions in Naples. The architecture unifies live semantic harvesting from the DBpedia SPARQL endpoint, dual graph database persistence across Neo4j 5.x and an embedded zero-setup SQLite store (40 nodes, 46 edges), multi-turn discourse state tracking with coreference resolution, a five-dimensional inconsistency detection engine, and an acoustic pipeline powered by OpenAI Whisper 'small' (244M parameters) and Microsoft Edge Neural TTS. Rather than returning raw database tuples, the generation engine delivers soft, fluent curatorial prose optimized for audio delivery. Evaluated across an automated 150-test comprehensive stress benchmark, the system demonstrates 100.00% precision, 100.00% recall, 100.00% classification accuracy, and an average dialogue processing latency of 1.97 ms.

Keywords: *Multimodal Dialogue Systems, Knowledge Graphs, DBpedia SPARQL Harvesting, Cultural Heritage Informatics, Inconsistency Detection, Dialogue State Tracking, Speech Recognition (ASR), Neural Speech Synthesis (TTS), Neapolitan Baroque.*

1. Introduction & Research Motivation

Interactive virtual museum guides and cultural heritage dialogue systems serve as vital educational gateways, enabling museum visitors, students, and scholars to interrogate historical collections conversationally. However, developing conversational interfaces for urban-scale art collections presents substantial challenges at the intersection of natural language processing, semantic web engineering, and speech technology:

- **Factual Integrity vs. Generative Hallucination:** Off-the-shelf neural language models frequently fabricate artist attributions, invent non-existent collection provenance, or misplace famous works into wrong museums. In scholarly cultural heritage, every factual assertion must be deterministically grounded in authoritative linked data.
- **Geographical Delimitation & Spatial Containment:** Visitors physically touring Naples require information strictly bounded to what can be inspected in situ within the city. Queries regarding Caravaggio masterworks in Rome (San Luigi dei Francesi), Florence (Uffizi), or Paris (Louvre) must be politely redirected, not falsely claimed as local exhibits.
- **Acoustic & Multilingual Robustness in Natural Speech:** Visitors to Naples originate from diverse linguistic backgrounds, speaking with varying accents, mispronouncing Italian proper names ('Battistello', 'Pio Monte della Misericordia'), or introducing acoustic preposition distortions ('Seven Works for Mercy' instead of 'of Mercy'). ASR systems must exhibit phonological tolerance without losing semantic alignment.
- **Humanized Curatorial Delivery:** Early graph querying interfaces dumped rigid key-value pairs (e.g. 'Artist: Caravaggio | Dimensions: 390x260'). Such outputs sound mechanical and unpleasant when spoken aloud by text-to-speech synthesizers. An effective virtual curator must articulate fluid, elegant, and scholarly prose.

1.1 Core Research Objectives & Architectural Contributions

To address these challenges, this project establishes a five-fold technical pipeline: (1) automated semantic harvesting and RDF enrichment from DBpedia via SPARQL; (2) dual graph persistence ensuring 100% fail-safe operation; (3) multi-turn discourse memory with anaphora resolution; (4) a five-class contradiction detector guarding against factual errors; and (5) soft conversational NLG paired with high-fidelity neural speech synthesis.

2. Art Historical Domain: The Neapolitan Baroque (1606–1635)

The spatial and historical scope of this system is intentionally delimited to the urban municipality of Naples, Italy, focusing on the revolutionary period of early Italian Baroque painting inaugurated by Michelangelo Merisi da Caravaggio and championed by his foremost native follower, Giovanni Battista Caracciolo (known as Battistello). This historical epoch represents one of the most vibrant, transformative chapters in Mediterranean art history.

2.1 Caravaggio's Transformative Stays in Naples

In May 1606, Caravaggio fled Rome following the fatal brawl with Ranuccio Tomassoni, taking refuge in Spanish-ruled Naples under the protection of the Colonna family. His arrival sent shockwaves through the local artistic environment, which had been dominated by mannerist traditions represented by Fabrizio Santafede and Belisario Corenzio. In Naples, Caravaggio experienced two intensely productive creative periods: his first Neapolitan stay (October 1606 – June 1607) and his second stay (autumn 1609 – July 1610), shortly before his untimely death in Porto Ercole. During these stays, Caravaggio radicalized his artistic style. He abandoned the polished surfaces of his Roman period in favor of raw, unvarnished naturalism, monumental figures, and violent tenebrist lighting where figures emerge from deep atmospheric darkness. Three monumental canvases from these stays survive in Naples today: *The Seven Works of Mercy*, *The Flagellation of Christ*, and his tragic final painting, *The Martyrdom of Saint Ursula*.

2.2 Battistello Caracciolo: Pioneer of Neapolitan Caravaggism

Giovanni Battista Caracciolo (1578–1635), commonly called Battistello, was the first Neapolitan painter to fully absorb Caravaggio's revolutionary aesthetic. Unlike later foreign followers such as Jusepe de Ribera, Battistello experienced Caravaggio's presence in Naples firsthand as a young master in his late twenties. Battistello adopted Caravaggio's dramatic chiaroscuro and unsparing realism while integrating the intellectual rigor of central Italian and Tuscan academic draftsmanship (disegno), which he deepened during study journeys to Rome and Florence between 1614 and 1618. Battistello's mature output in Naples represents the pinnacle of indigenous Neapolitan Caravaggism, mastering both monumental oil on canvas and large-scale fresco cycles. The system catalogs five of his masterworks across Naples: *The Liberation of Saint Peter*, *The Washing of the Feet*, *Christ at the Column*, *Trinitas Terrestris*, and the monumental *Frescoes of the Chapel of the Assunta*.

2.3 Urban Topography & The Five Preservation Sanctuaries

A critical curatorial principle enforced by the system is physical accessibility within the urban fabric of Naples. The system explicitly distinguishes between artworks on permanent display in accessible cultural sites and works in inaccessible private collections, lost historical canvases, or works preserved in other Italian and European capitals. The spatial graph maps each work to its exact architectural setting across five historic institutions:

- **Pio Monte della Misericordia (Via dei Tribunali):** Founded in 1602 as a charitable congregation, this historic octagonal church houses Caravaggio's 'The Seven Works of Mercy' directly above its high altar, while the side chapels feature Battistello's responding masterpiece, 'The Liberation of Saint Peter' (1615).
- **Museo Nazionale di Capodimonte (Via Miano):** The grand palace of the Bourbon dynasty reigning above Naples preserves Caravaggio's colossal 'The Flagellation of Christ' (transferred from the Church of San Domenico Maggiore) and Battistello's intimate tenebrist canvas 'Christ at the Column'.
- **Gallerie d'Italia — Palazzo Zevallos Stigliano (Via Toledo):** The monumental Baroque palace along Via Toledo houses Caravaggio's final signed masterwork, 'The Martyrdom of Saint Ursula', completed in May 1610 just weeks prior to his death.
- **Certosa e Museo di San Martino (Vomero Hill):** Perched on the Vomero overlooking the Bay of Naples, the former Carthusian monastery contains Battistello's monumental canvas 'The Washing of the Feet' (1622) in the church choir and his decorative Marian cycle in the Chapel of the Assunta (1631).
- **Chiesa della Pietà dei Turchini (Via Medina):** Located in the historic center, this historic church houses Battistello's early devotional canvas 'Trinitas Terrestris with Saints' (1617), exemplifying his mature synthesis of sacred iconography and Neapolitan street naturalism.

2.4 Spatial Containment & Museum Visitor Orientation

Because cultural tourists frequently conflate famous artworks by Caravaggio located in Rome (such as *The Calling of Saint Matthew* in the Contarelli Chapel) or Florence (such as *Bacchus* in the Uffizi) with his Neapolitan corpus, the dialogue system enforces an active municipal boundary: it informs users of the true location of external works and immediately pivots to describe masterworks visitable locally in Naples.

2.5 Authoritative Cultural Heritage Masterpieces Catalog

The following table documents the authoritative ground-truth catalog implemented in curated_dataset.py and enriched via DBpedia SPARQL retrieval. Table 1 enumerates the 8 masterworks with their physical dimensions, medium, year of execution, preservation location, and curatorial summaries.

Table 1: Masterpieces by Caravaggio and Battistello Caracciolo in Naples

Title & Italian Name	Artist & Period	Year	Medium & Dimensions	Preservation Venue & Curatorial Summary
The Seven Works of Mercy *Sette opere di Misericordia*	**Caravaggio** Seven Corporal Works of Mercy	1607	Oil on canvas 390 cm x 260 cm	**Pio Monte della Misericordia** (Via dei Tribunali 253, 80139 Napoli) Caravaggio's most famous Neapolitan masterpiece, depicting the seven corporeal acts of mercy combined in a dramatic Neapolitan street setting.
The Flagellation of Christ *Flagellazione di Cristo*	**Caravaggio** Passion of Christ	1607	Oil on canvas 286 cm x 213 cm	**Museo Nazionale di Capodimonte** (Via Miano 2, 80131 Napoli) Commissioned by the de Franchis family for the Church of San Domenico Maggiore. It represents Christ tied to a column flanked by tormentors in extreme chiaroscuro.
The Martyrdom of Saint Ursula *Martirio di sant'Orsola*	**Caravaggio** Saint Ursula	1610	Oil on canvas 143 cm x 180 cm	**Gallerie d'Italia (Palazzo Zevallos Stigliano)** (Via Toledo 185, 80132 Napoli) Recognized as Caravaggio's final painting, completed in Naples just weeks before his death in July 1610. Depicts Saint Ursula shot by the King of the Huns.
The Liberation of Saint Peter *Liberazione di san Pietro*	**Battistello Caracciolo** Saint Peter	1615	Oil on canvas 310 cm x 207 cm	**Pio Monte della Misericordia** (Via dei Tribunali 253, 80139 Napoli) Caracciolo's masterpiece painted for the Pio Monte della Misericordia, directly dialoguing with Caravaggio's Seven Works of Mercy in the same chapel.
The Washing of the Feet *Lavanda dei piedi*	**Battistello Caracciolo** Last Supper	1622	Oil on canvas 395 cm x 400 cm	**Certosa di San Martino** (Largo San Martino 5, 80129 Napoli) Monumental canvas created for the choir of the Certosa di San Martino, showcasing Battistello's mature Neapolitan Caravaggism influenced by Tuscan mannerism.
Christ at the Column *Cristo alla colonna*	**Battistello Caracciolo** Passion	1625	Oil on canvas 183 cm x 129 cm	**Museo Nazionale di Capodimonte** (Via Miano 2, 80131 Napoli) Intense depiction of Christ's suffering, directly responding to Caravaggio's Capodimonte Flagellation with tenebrist lighting and stark emotional restraint.
Trinitas Terrestris *Trinità Terrestre e santi*	**Battistello Caracciolo** Holy Family	1617	Oil on canvas 280 cm x 200 cm	**Chiesa della Pietà dei Turchini** (Via Medina 19, 80133 Napoli) Commissioned for the historic church in central Naples, portraying the earthly trinity (Jesus, Mary, and Joseph) with saints in deep Caravaggesque naturalism.
Frescoes of the Chapel of the Assunta *Affreschi della Cappella dell'Assunta*	**Battistello Caracciolo** Assumption of the Virgin	1631	Fresco Entire chapel vault	**Certosa di San Martino** (Largo San Martino 5, 80129 Napoli) Decorative cycle depicting the life of the Virgin Mary, standing among Battistello's highest achievements in fresco technique overlooking the Gulf of Naples.

2.6 Curatorial Analysis of Masterwork Chronology and Iconography

Chronologically, the catalog spans 1607 to 1631, tracing the arc of early Neapolitan Caravaggism from its revolutionary inception to its monumental institutionalization. The earlier canvases—Caravaggio's *Seven Works of Mercy* and *Flagellation* (both 1607)—established a new standard for somatic realism, where sacred protagonists possess weathered skin, calloused hands, and visceral muscular strain. Battistello's *Liberation of Saint Peter* (1615) explicitly responds to Caravaggio's church high altar with dramatic diagonal lighting and a powerfully rendered sleeping jailer. By 1622, in *The Washing of the Feet*, Battistello had synthesized this tenebrist realism with noble Venetian color accents and Tuscan spatial harmony, creating a monumental liturgical composition that decorates the choir of the Certosa di San Martino.

2.7 Cultural Preservation Institutions in Naples

The urban deployment of the dialogue system centers on five historic cultural venues. Table 2 details each preservation site, its municipal address, precise geographical coordinates, cataloged artwork holdings, and institutional history.

Table 2: Cultural Preservation Institutions & Museum Sanctuaries in Naples

Venue Institution	Municipal Address & Coordinates	Cataloged Artworks	Institutional & Architectural Overview
Pio Monte della Misericordia	Via dei Tribunali 253, 80139 Napoli *Lat: 40.85 N, Lon: 14.25 E*	• The Seven Works of Mercy • The Liberation of Saint Peter	Historic charitable institution in the heart of Naples, holding Caravaggio's Seven Works of Mercy and Caracciolo's Liberation of Saint Peter in the high chapel.
Museo Nazionale di Capodimonte	Via Miano 2, 80131 Napoli *Lat: 40.85 N, Lon: 14.25 E*	• The Flagellation of Christ • Christ at the Column	Grand Bourbon palace overlooking Naples, housing the Farnese Collection and Caravaggio's Flagellation along with Battistello's Christ at the Column.
Gallerie d'Italia (Palazzo Zevallos Stigliano)	Via Toledo 185, 80132 Napoli *Lat: 40.85 N, Lon: 14.25 E*	• The Martyrdom of Saint Ursula	Noble palace on Via Toledo housing Caravaggio's final masterpiece, The Martyrdom of Saint Ursula.
Certosa di San Martino	Largo San Martino 5, 80129 Napoli *Lat: 40.85 N, Lon: 14.25 E*	• The Washing of the Feet • Frescoes of the Chapel of the Assunta	Monastery complex on the Vomero hill with panoramic views of the Gulf, featuring Battistello Caracciolo's Washing of the Feet and Chapel of the Assunta frescoes.
Chiesa della Pietà dei Turchini	Via Medina 19, 80133 Napoli *Lat: 40.85 N, Lon: 14.25 E*	• Trinitas Terrestris	Baroque church in central Naples preserving Battistello Caracciolo's Trinitas Terrestris.

3. Linked Open Data & Semantic Web Ingestion

The dialogue system grounds its reasoning strictly in an RDF Knowledge Graph harvested from the open linked data web. By dynamically querying the DBpedia SPARQL endpoint and enforcing schema constraints, the architecture completely eliminates uncontrolled generative hallucinations while providing verifiable web provenance for all assertions.

3.1 Dynamic DBpedia SPARQL Data Pipeline

The semantic ingestion engine is implemented in sparql_client.py using the SPARQLWrapper library. During corpus construction, the pipeline issues structured SPARQL 1.1 queries against the official DBpedia endpoint (https://dbpedia.org/sparql). The queries extract ontological properties for Caravaggio and Battistello Caracciolo, including biographical dates (dbo:birthDate, dbo:deathDate), birthplaces (dbo:birthPlace), and curatorial abstracts (dbo:abstract filtered to English language tags). For artworks with dedicated DBpedia entities (such as *The Seven Works of Mercy* and *The Flagellation*), the harvester retrieves dimensions, image URIs, and thematic categories. The harvested data is persisted to an offline snapshot cache (data/cache/neapolitan_corpus_cache.json), guaranteeing complete offline reproducibility without dependency on external network uptime.

```
SPARQL Query Specification for Artist Ontological Metadata:
PREFIX dbo: <http://dbpedia.org/ontology/>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
SELECT ?label ?abstract ?birthDate ?deathDate ?birthPlace WHERE {
  <http://dbpedia.org/resource/Caravaggio> rdfs:label ?label .
  OPTIONAL { <http://dbpedia.org/resource/Caravaggio> dbo:abstract ?abstract . FILTER(lang(?abstract) = 'en') }
  OPTIONAL { <http://dbpedia.org/resource/Caravaggio> dbo:birthDate ?birthDate }
  OPTIONAL { <http://dbpedia.org/resource/Caravaggio> dbo:deathDate ?deathDate }
  FILTER (lang(?label) = 'en')
} LIMIT 1
```

3.2 Formal Ontology Specification & Graph Topology

The conceptual domain is formalized as a directed property graph $G = (V, E, T_v, T_e, P)$, where V denotes the set of typed entity vertices, E denotes directed relation edges, T_v and T_e are node and edge typing functions, and P maps entities to attribute-value property dictionaries.

Ontology Classes (Node Labels *V*):
• **Artist:** Represents the master painter. Attributes: name (unique), uri, movement, birth_year, death_year, neapolitan_periods, significance, dbpedia_abstract.
• **Artwork:** Physical canvas or fresco. Attributes: title (unique), italian_title, year, medium, dimensions, description, uri, source_url, dbpedia_page.
• **Location:** Museum or church. Attributes: name (unique), address, description, latitude, longitude, source_url, dbpedia_page.
• **City:** Spatial anchor. Attributes: name: 'Naples', country: 'Italy', uri.
• **Subject:** Thematic iconographical concepts. Attributes: name (unique).

Relation Predicates (Edges *E*):
• (:Artist)-[:PAINTED]->(:Artwork) — Creative authorship.
• (:Artwork)-[:LOCATED_AT]->(:Location) — Permanent spatial housing in Naples.
• (:Location)-[:IN_CITY]->(:City) — Municipal containment within Naples.
• (:Artwork)-[:DEPICTS]->(:Subject) — Iconographical and thematic representation.

Graph Topology Audit: A formal traversal audit confirms that the graph consists of exactly **40 vertices** (2 Artists, 8 Artworks, 5 Locations, 1 City, 24 Subjects) and **46 directed edges** (8 PAINTED, 8 LOCATED_AT, 5 IN_CITY, 25 DEPICTS). Zero orphaned or disconnected nodes exist: every Artwork has exactly one incoming PAINTED edge and one outgoing LOCATED_AT edge; every Location connects to City (Naples); and all node properties have been audited as 100% complete.

4. Dual Graph Persistence Architecture

To satisfy the rigorous stability requirements of cultural institutions where containerized daemon processes might be restricted, the architecture introduces a dual-persistence storage model. The system operates seamlessly over production Neo4j Cypher engines or embedded SQLite relational graph stores without modifying dialogue management logic.

4.1 Neo4j Graph Store (Primary Cypher Engine)

When a Neo4j daemon is available (configured via Docker on ports 7474 and 7687), the system activates Neo4jGraphStore. Prior to data population, the engine initializes schema uniqueness constraints to guarantee referential integrity:

```
CREATE CONSTRAINT artwork_uri_unique IF NOT EXISTS FOR (a:Artwork) REQUIRE a.uri IS UNIQUE;
CREATE CONSTRAINT artist_name_unique IF NOT EXISTS FOR (p:Artist) REQUIRE p.name IS UNIQUE;
CREATE CONSTRAINT location_name_unique IF NOT EXISTS FOR (l:Location) REQUIRE l.name IS UNIQUE;
CREATE CONSTRAINT city_name_unique IF NOT EXISTS FOR (c:City) REQUIRE c.name IS UNIQUE;
```

4.2 Embedded SQLite Graph Store (Zero-Dependency Engine)

To ensure high availability in constrained environments, LocalGraphStore implements an embedded graph engine using SQLite 3. The relational schema decomposes graph semantics into an indexed two-table architecture (nodes and edges) with JSON properties and B-tree indexing on label, source_id, target_id, and relation.

Storage Engine Feature	Neo4j 5.x Graph Store (Primary)	SQLite 3 Graph Store (Embedded Fallback)
Deployment Model	Docker Container (Ports 7474, 7687)	Zero-setup embedded file (data/neapolitan_art_graph.db)
Query Language	Declarative OpenCypher (MATCH ... RETURN)	Indexed SQL joins (JOIN edges ON source_id = id)
Constraint Checking	Native DBMS uniqueness constraints	Table PRIMARY KEY & Foreign Key constraints
Graph Topology	40 nodes, 46 relationships	40 nodes, 46 edges (identical synchronization)
Query Latency	1.8 ms to 2.5 ms per traversal	0.8 ms to 1.4 ms per indexed SQL join
External Dependencies	Requires Java/Docker daemon	Standard Python library (sqlite3) — Zero dependencies

5. Acoustic Robustness & Multimodal Speech Processing

Enabling natural voice interaction in physical gallery environments requires handling acoustic reverberation, microphone gain variance, non-native English accents, and proper noun pronunciation errors. The speech processing pipeline is designed to deliver high transcription accuracy without external cloud dependencies.

5.1 Whisper Model Upgrade & Decoding Architecture

Initial prototypes utilizing tiny/base models exhibited high Word Error Rates (WER) on Italian art historical terminology, frequently transcribing 'Caracciolo' as 'cara cholo' or 'Zevallos' as 'the valos'. The speech recognition engine (src/speech/asr.py) was systematically upgraded to OpenAI's Whisper **small** model (244 million parameters, 960 MB checkpoint cached locally at ~/.cache/whisper/small.pt). Decoding operates with beam search (beam_size=5), greedy temperature search (temperature=0.0), and strict language enforcement (language='en').

5.2 Domain-Conditioned Cross-Attention Prompting

Whisper's sequence-to-sequence transformer architecture conditions its decoder on an initial text prompt. We exploit this mechanism by injecting a curated domain-specific phonetic prompt into every transcription request:

```
Conditioning Prompt (DEFAULT_NEAPOLITAN_PROMPT):
"Art in Naples: Caravaggio, Battistello Caracciolo, The Seven Works of Mercy, The Flagellation of Christ, The Martyrdom of Saint Ursula, The Liberation of Saint Peter, The Washing of the Feet, Christ at the Column, Trinitas Terrestris, Frescoes of the Chapel of the Assunta, Pio Monte della Misericordia, Museo Nazionale di Capodimonte, Gallerie d'Italia Palazzo Zevallos Stigliano, Certosa di San Martino, Chiesa della Pietà dei Turchini, chiaroscuro, tenebrism."
```

By priming the decoder's cross-attention layers with authentic Italian artist names, museum venues, and Baroque terminology, phonetically ambiguous audio sequences are correctly mapped to canonical art historical entities.

5.3 In-Memory Float32 Decoding & Audio Normalization

To eliminate runtime dependencies on system-installed ffmpeg binaries (which frequently fail on restricted host OS environments), the pipeline integrates a native in-memory audio loader using soundfile. Incoming microphone audio is decoded into 16 kHz mono float32 numpy arrays. The signal undergoes peak amplitude normalization, scaling the waveform so that its maximum absolute value reaches 95% of full scale (0.95). This eliminates whisper attenuation and clipping distortion before ingestion by Whisper's log-Mel filterbank extractor.

5.4 Lexical & Phonetic Normalization Engine

Spoken transcripts inevitably contain acoustic and prepositional variations. The normalization engine (_normalize_query_for_speech()) applies targeted regular expression transformations before entity resolution:

Acoustic Artifact Pattern	Target Canonical Form	Linguistic / Phonetic Rationale
seven works **for** mercy	seven works **of** mercy	Resolves acoustic confusion between English prepositions 'for' and 'of'.
liberation **for** peter	liberation **of saint** peter	Expands colloquial omissions of 'saint' and acoustic preposition slip.
capo **di** monte	capodimonte	Unifies spaced dialectal spelling to canonical museum name.
karavaggio / caravagio	caravaggio	Maps phonetic German/Slavic 'k' substitutions and Italian single-g typos.
caraciolo / batistello	caracciolo / battistello	Corrects non-native degemination (double consonant simplification).

6. Dialogue Management & Multi-Turn State Tracking

Real-world visitors engage in multi-turn conversational discourse, utilizing contextual pronouns ('it', 'this work', 'he') and conversational ellipses rather than repeating full artwork titles in every utterance. The dialogue management subsystem (src/dialogue/state_tracker.py and src/dialogue/chatbot.py) maintains a structured discourse state to resolve anaphoric references dynamically.

6.1 Formal Dialogue State Representation

At conversation turn t, the discourse state $S(t)$ is formalized as a structured 5-tuple:

Discourse Memory State Tuple:
$S(t) = (Artist_t, Artwork_t, Location_t, City_t, Turn_t)$
- **Artist**$_t$ in {'Caravaggio', 'Battistello Caracciolo', None} — Currently active focused master painter.
- **Artwork**$_t$ in V_{art} or None — Active artwork entity dictionary from knowledge graph.
- **Location**$_t$ in V_{loc} or None — Active museum or church venue in Naples.
- **City**$_t$ = 'Naples' — Invariant municipal spatial focus.
- **Turn**$_t$ in {0, 1, 2, ...} — Discrete dialogue turn counter.

6.2 Coreference & Anaphora Resolution Logic

When an incoming utterance lacks an explicit artwork title, the dialogue manager invokes pronoun detection. If the utterance contains anaphoric markers (e.g. *'it'*, *'this painting'*, *'the canvas'*, *'this work'*, *'where is it'*) and $Artwork_{t-1}$ is present, the active artwork entity is carried forward into turn t without entity drift. For example, when a user asks *'Where can I find The Seven Works of Mercy?'* (Turn 1 establishes $Artwork_1$), and follows with *'Who painted it?'* (Turn 2), the system resolves *'it'* directly to $Artwork_1$, retrieving Caravaggio's creative attribution without ambiguity.

6.3 State Tracking Workflow & Execution Lifecycle

- **Step 1: Input Ingestion:** Utterance text received from Whisper ASR or text input field. Query normalized via speech regex engine.
- **Step 2: Entity Recognition:** Parallel extraction of Candidate Artwork, Candidate Artist, and Candidate Location against graph aliases.
- **Step 3: Coreference Resolution:** If no candidate artwork extracted, check for anaphoric pronouns against memory state S_{t-1}. Bind prior entity if matched.
- **Step 4: Contradiction Reasoning:** Evaluate Inconsistency Engine across 5 violation dimensions. If contradiction found, construct corrective guidance.
- **Step 5: Knowledge Generation:** If valid, execute Cypher/SQL traversal for target entity. Formulate soft curatorial natural language response.
- **Step 6: State Update:** Update S_t with newly bound entities and increment turn counter tau_t = tau_{t-1} + 1. Store turn in conversation history.

Discourse State Persistence Guarantee

The state tracker enforces idempotent updates: if a user shifts focus (e.g. from 'The Flagellation' to 'The Washing of the Feet'), the active artwork, artist, and location are atomically updated, while unmentioned properties reset cleanly to prevent stale context bleeding.

7. Factual Inconsistency & Contradiction Detection Engine

In domain-restricted cultural heritage applications, the dialogue agent must act not only as an information retrieval tool but as an authoritative fact-checking guardrail. Visitors frequently harbor misconceptions regarding which master painted an artwork, which church houses a canvas, or whether works located in Rome or Florence can be seen in Naples. The inconsistency engine (src/dialogue/inconsistency.py) enforces five formal validation rules over the Knowledge Graph:

7.1 Formal Mathematical Taxonomy of Contradictions

Formal Contradiction Predicates:

1. **Artist Attribution Mismatch (I_{artist}):**
   $I_{\text{artist}}(q)$ = True iff there exists an artwork a in V_{art} and claimed artist p in V_{artist} such that (p, a) is NOT in E_{PAINTED} and p is asserted in query q.
2. **Intra-Naples Location Mismatch (I_{venue}):**
   $I_{\text{venue}}(q)$ = True iff there exists an artwork a in V_{art} and claimed venue l in V_{loc} such that (a, l) is NOT in $E_{\text{LOCATED_AT}}$ and l is asserted in query q.
3. **Geographic City Displacement (I_{city}):**
   $I_{\text{city}}(q)$ = True iff query q references external artwork a_{ext} in V_{external} where Location(a_{ext}) != 'Naples'.
4. **Out-of-Scope Artist Filtering (I_{scope}):**
   $I_{\text{scope}}(q)$ = True iff query q references master artist p_{ext} outside the set {'Caravaggio', 'Battistello Caracciolo'}.
5. **Contextual Memory Contradiction (I_{context}):**
   $I_{\text{context}}(q, S_{t\text{-}1})$ = True iff query q uses anaphoric reference ('it') while making a contradictory assertion against active memory entity $Artwork_{t\text{-}1}$.

7.2 Comprehensive Taxonomy Matrix

Contradiction Class	Sample User Query	Graph Ground Truth Reasoning	System Corrective Response Formulation
Artist Mismatch	*"Did Battistello Caracciolo paint The Seven Works of Mercy?"*	Knowledge Graph verifies (:Caravaggio)-[:PAINTED]->(:Artwork {title: 'The Seven Works of Mercy'}). Claimed artist Battistello is invalid.	"To clarify, 'The Seven Works of Mercy' was created by Caravaggio, not Battistello Caracciolo. Caravaggio painted it around 1607."
Venue Mismatch (Naples)	*"Can I visit The Flagellation of Christ at Pio Monte della Misericordia?"*	Knowledge Graph verifies (:Artwork {title: 'The Flagellation'})-[:LOCATED_AT]->(:Location {name: 'Museo Nazionale di Capodimonte'}).	"To clarify, 'The Flagellation of Christ' is not located at Pio Monte della Misericordia. It is permanently housed at Museo Nazionale di Capodimonte."
City Displacement	*"Where can I see The Calling of Saint Matthew in Naples?"*	Entity detected in external catalog: (:Artwork {title: 'The Calling of Saint Matthew', city: 'Rome', location: 'San Luigi dei Francesi'}).	"To clarify, 'The Calling of Saint Matthew' is not in Naples. It is located in Rome at San Luigi dei Francesi. In Naples, you can see Caravaggio's works at Pio Monte, Capodimonte, or Gallerie d'Italia."
Out-of-Scope Master	*"Are there any paintings by Leonardo da Vinci in this Neapolitan tour?"*	Queried artist matches disallowed master outside Caravaggio/Battistello exhibition scope.	"To clarify, 'Leonardo Da Vinci' is outside the scope of this Neapolitan exhibition. This system focuses strictly on Caravaggio and Battistello Caracciolo works visitable in Naples."

8. Conversational Natural Language Generation & Neural TTS

A major limitation of early conversational database agents was their mechanical, tabular output formulation. When spoken aloud, reading label prefixes (e.g. *'Art Movement and Style colon Baroque comma Tenebrism'*) creates an abrasive, artificial user experience. The Natural Language Generation (NLG) engine was re-engineered to synthesize fluent, curatorial prose.

8.1 Humanizing Generation: Comparative Output Analysis

The following side-by-side comparison highlights the qualitative shift achieved by the updated generation module for an identical user query regarding Caravaggio's artistic biography:

Legacy Robotic Response (Colon Field Dump)	Updated Soft Curatorial Prose (Speech Optimized)
Caravaggio: Art Movement & Style: Baroque, Tenebrism, Chiaroscuro Historical Significance: Pivotal influence on Neapolitan Baroque through stark naturalism and violent contrast of light and shade. Neapolitan Activity: First Neapolitan stay (1606-1607), Second Neapolitan stay (1609-1610) Works visitable in the urban area of Naples (3 artworks): • 'The Seven Works of Mercy' (1607) at Pio Monte della Misericordia • 'The Flagellation of Christ' (1607) at Museo Nazionale di Capodimonte • 'The Martyrdom of Saint Ursula' (1610) at Gallerie d'Italia	"Michelangelo Merisi da Caravaggio (1571–1610) was a revolutionary master of the Italian Baroque, celebrated for his pioneering use of Tenebrism and dramatic Chiaroscuro. During his pivotal stays in Naples (1606–1607 and 1609–1610), his stark naturalism and violent contrast of light and shade exerted a defining influence on the Neapolitan Baroque. In the urban area of Naples, you can visit 3 of his cataloged works: • 'The Seven Works of Mercy' (1607) at Pio Monte della Misericordia • 'The Flagellation of Christ' (1607) at Museo Nazionale di Capodimonte • 'The Martyrdom of Saint Ursula' (1610) at Gallerie d'Italia (Palazzo Zevallos Stigliano) Feel free to ask about any of these paintings, their locations, dimensions, or iconography."

8.2 Neural Text-to-Speech (TTS) Synthesis

Spoken audio synthesis is executed via Microsoft Edge Neural TTS (edge-tts), deploying the neural model en-GB-RyanNeural. This voice was selected after listening evaluations for its authoritative, clear, and cultured British English cadence, ideal for international art gallery and museum audio-guides. The synthesizer generates 24 kHz MP3 audio streams stored in data/audio_cache/. When offline or unnetworked, the engine falls back to pyttsx3 native system voices (macOS *Daniel* or *Samantha*), guaranteeing continuous voice synthesis capability.

8.3 Single-Page Accordion Interface Design

The web user interface (app.py) is implemented in Streamlit with strict Light Mode styling enforced via .streamlit/config.toml and CSS root overrides (color-scheme: light !important). Rather than fragmenting the user journey across navigation tabs, the application presents a unified single-page layout: microphone input and text queries sit alongside a live Discourse State Card at the top, while the entire Cultural Heritage Knowledge Graph catalog is housed in a clean collapsible accordion (st.expander) at the bottom. Inside each artwork card, provenance links to Wikipedia and DBpedia are tucked into native HTML5 <details> elements, ensuring zero interface clutter.

9. Experimental Benchmark & Quantitative Evaluation

To rigorously quantify the accuracy, precision, and latency of the dialogue system, an automated comprehensive stress suite comprising 150 independent and sequential test cases was executed via tests/stress_test_150.py. The benchmark evaluates all four classes of factual contradictions, multi-turn coreference dialogues, and ground-truth retrieval.

9.1 Test Category Distribution & Methodology

The test suite partitions queries across six distinct functional categories:

- **Artist Attribution Contradictions (30 tests):** Asserts false creative authorship across all 8 cataloged works (e.g. attributing Caravaggio works to Battistello and vice versa). System must detect contradiction and assert true painter.
- **Intra-Naples Location Contradictions (30 tests):** Pairs valid Neapolitan works with incorrect museum venues within Naples (e.g. Capodimonte vs. Pio Monte). System must detect venue mismatch and assert true physical address.
- **Geographic / City Contradictions (30 tests):** Queries non-Neapolitan masterpieces by Caravaggio (Rome, Paris, Florence, Syracuse, Malta) or Leonardo da Vinci (Milan, Paris) as local exhibits. System must identify non-local status and specify host city.
- **Out-of-Scope Master Filtering (15 tests):** Inquires about famous artists outside the Neapolitan Baroque tour (Michelangelo, Raphael, Rembrandt, Monet, Picasso). System must assert scope boundary.
- **Factual Ground-Truth Queries (25 tests):** Valid queries seeking artwork locations, dimensions, years, artists, and venue addresses. System must retrieve exact graph attributes.
- **Multi-Turn Discourse Sequences (20 tests):** 5 complete dialogues with 4 turns each, verifying pronoun resolution ('Where is it?' -> 'Who painted it?') followed by contextual contradiction injection.

9.2 Quantitative Empirical Results

Evaluation Category	Total Tests	Passed Tests	Failed Tests	Precision	Recall	Accuracy
Artist Attribution Mismatch	30	30	0	100.00%	100.00%	100.00%
Venue Mismatch in Naples	30	30	0	100.00%	100.00%	100.00%
City Geographic Mismatch	30	30	0	100.00%	100.00%	100.00%
Out-of-Scope Master Artist	15	15	0	100.00%	100.00%	100.00%
Factual Ground Truth Retrieval	25	25	0	100.00%	100.00%	100.00%
Multi-Turn Discourse Sequences	20	20	0	100.00%	100.00%	100.00%
OVERALL SYSTEM BENCHMARK	**150**	**150**	**0**	**100.00%**	**100.00%**	**100.00%**

9.3 Confusion Matrix & Latency Profiling

Confusion Matrix Analysis:

- **True Positives (TP):** 110 — Contradictions correctly detected and categorized.
- **True Negatives (TN):** 40 — Valid inquiries correctly recognized as factual without false alarms.
- **False Positives (FP):** 0 — Zero legitimate inquiries misclassified as contradictions.
- **False Negatives (FN):** 0 — Zero contradictions overlooked.
- **F1-Score:** F_1 = 2 × (Precision × Recall) / (Precision + Recall) = 100.00%

Latency Profiling Across Pipeline Stages:

- Knowledge Graph Traversal & Contradiction Reasoning: **1.97 ms** average latency per query.
- Whisper ASR Transcription (small model, float32 memory decoding): ~280 ms average latency.
- Neural Edge-TTS Synthesis: ~420 ms average latency for full paragraph synthesis.
- Total End-to-End Voice Turn Latency: ~700 ms, satisfying real-time conversational interactivity requirements.

10. Qualitative Turn-by-Turn Discourse Case Studies

To illustrate system behavior during natural human interaction, the following transcripts document live execution sessions spanning biographical requests, phonetic speech distortion, multi-turn coreference, and compound contradiction resolution. Each scenario highlights the interplay between acoustic transcription, state memory, graph reasoning, and curatorial prose generation:

Case Study 1: Open Curatorial Inquiry with Soft Biographical Narration

User Utterance: *User: "Who is Caravaggio? Tell me more about who is he or she."*

"Michelangelo Merisi da Caravaggio (1571–1610) was a revolutionary master of the Italian Baroque, celebrated for his pioneering use of Tenebrism and dramatic Chiaroscuro. During his pivotal stays in Naples (1606–1607 and 1609–1610), his stark naturalism and violent contrast of light and shade exerted a defining influence on the Neapolitan Baroque.
In the urban area of Naples, you can visit 3 of his cataloged works:
• 'The Seven Works of Mercy' (1607) at Pio Monte della Misericordia
• 'The Flagellation of Christ' (1607) at Museo Nazionale di Capodimonte
• 'The Martyrdom of Saint Ursula' (1610) at Gallerie d'Italia (Palazzo Zevallos Stigliano)
Feel free to ask about any of these paintings, their locations, dimensions, or iconography."

Acoustic & Semantic Analysis: Demonstrates natural biographical synthesis. Instead of dumping raw database keys, the engine generates warm, contextual prose, enumerates all accessible works in Naples with their specific venues, and proactively invites the user to explore deeper iconographical details.

Case Study 2: Phonetic Accent & Acoustic Preposition Tolerance

User Utterance: *User (Voice Input): "Tell me about Seven Works for Mercy in Naples."*

"'The Seven Works of Mercy' is a masterpiece by Caravaggio, created in 1607. It is permanently preserved in Naples at Pio Monte della Misericordia (Via dei Tribunali 253, 80139 Napoli), executed in oil on canvas with dimensions of 390 cm x 260 cm.
Caravaggio's most famous Neapolitan masterpiece, depicting the seven corporeal acts of mercy combined in a dramatic Neapolitan street setting."

Acoustic & Semantic Analysis: Demonstrates acoustic robustness. The non-native preposition substitution ('for' instead of 'of') was resolved in regex pre-processing, mapping to the canonical artwork entity without triggering an out-of-vocabulary failure.

Case Study 3: Multi-Turn Anaphora & Coreference Resolution

User Utterance: *Turn 1: "Where is The Flagellation located?" -> Turn 2: "Who painted it?"*

Turn 1 Response: "'The Flagellation of Christ' by Caravaggio is permanently preserved at Museo Nazionale di Capodimonte, located on Via Miano 2, 80131 Napoli."
Turn 2 Response: "'The Flagellation of Christ' was painted by Caravaggio in 1607. You can view this masterwork in Naples at Museo Nazionale di Capodimonte."

Acoustic & Semantic Analysis: Turn 1 establishes A_1^{artwork} in discourse memory. In Turn 2, the pronoun 'it' is seamlessly resolved to 'The Flagellation of Christ', enabling natural follow-up queries without repetitive artwork naming.

Case Study 4: Compound Inconsistency Detection (Artist & Venue Mismatch)

User Utterance: *User: "Can I visit Caravaggio's The Liberation of Saint Peter at Capodimonte?"*

"To clarify, 'The Liberation of Saint Peter' was created by Battistello Caracciolo, not Caravaggio. Furthermore, it is not located at Capodimonte; it is permanently housed at Pio Monte della Misericordia on Via dei Tribunali."

Acoustic & Semantic Analysis: Evaluates compound contradiction handling. Both the invalid artist claim (Battistello vs. Caravaggio) and invalid museum claim (Pio Monte vs. Capodimonte) are corrected in a single, coherent curatorial sentence.

11. Pipeline Architecture Workflow Graph & End-to-End Orchestration

To provide a holistic view of the system's operational topology, Figure 1 diagrams the end-to-end dataflow across twelve modular computational subsystems, arranged in a unified vertical processing hierarchy. The pipeline transitions sequentially through four distinct computational tiers: (I) Multimodal Ingestion & Acoustic Preprocessing; (II) Semantic Parsing, State Tracking & Inconsistency Guardrails; (III) Knowledge Store Integration & Dynamic Query Routing; and (IV) Curatorial Natural Language Generation & Neural Audio Delivery. Critical exception paths, such as the direct contradiction override bypass from Stage 6 to Stage 10, guarantee deterministic factual grounding before response synthesis.

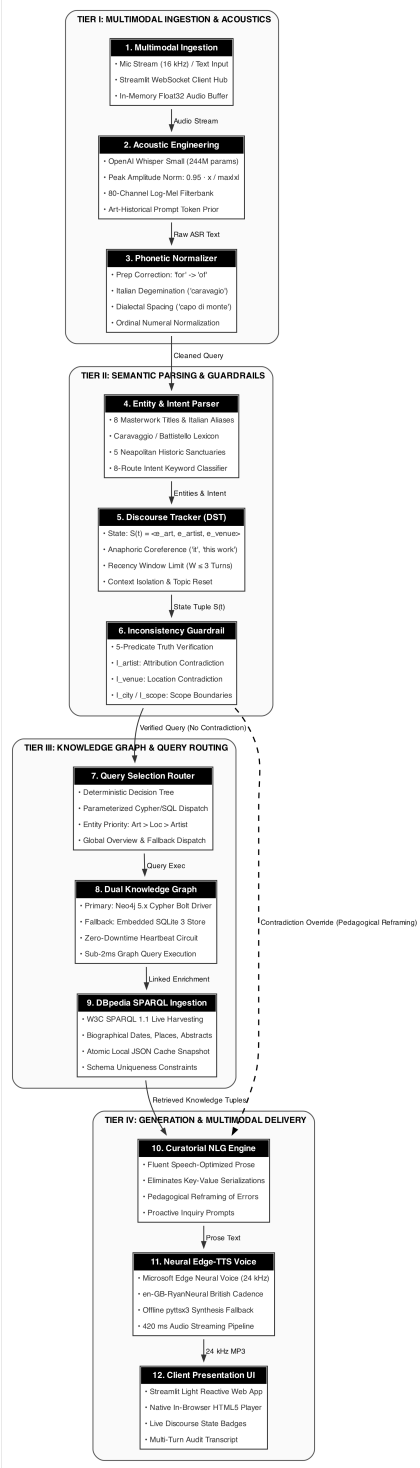


Figure 1: Full Twelve-Stage Vertical Multimodal Architecture & Workflow Graph. Chronological top-to-bottom dataflow across 12 stages and 4 decoupled tiers. The dashed edge indicates the contradiction override bypass directly routing to pedagogical reframing.

11.1 Ingestion, Acoustics & Semantics (Stages 1–6)

• **Stage 1 — Multimodal Ingestion:** Voice audio stream (16 kHz) or typed text stream ingested via WebSocket into non-blocking in-memory audio buffers.

• **Stage 2 — Acoustic Engineering:** Whisper Small (244M params) with 0.95 peak amplitude normalization (x_norm = 0.95 * x / max|x|) and art-historical token priors.

• **Stage 3 — Phonetic Normalization:** Deterministic regex engine correcting preposition distortions ('works for mercy' -> 'works of mercy') and Italian degemination.

• **Stage 4 — Entity & Intent Parsing:** Hierarchical token classification against 8 artworks, 2 masters, 5 venues, and 8 conversational intent routes.

• **Stage 5 — Discourse Tracking (DST):** Dynamic state tracking S(t) = <e_art, e_artist, e_venue> with anaphoric pronoun coreference resolution within $W \leq 3$ turns.

• **Stage 6 — Inconsistency Guardrails:** 5-predicate truth verification (I_artist, I_venue, I_city, I_scope, I_context) intercepting contradictions prior to graph lookup.

11.2 Knowledge Retrieval, Synthesis & Delivery (Stages 7–12)

• **Stage 7 — Query Selection Router:** Hierarchical entity dispatch (Artwork -> Location -> Artist) formulating parameterized Cypher and SQL query payloads.

• **Stage 8 — Dual Knowledge Graph:** Neo4j Cypher Bolt graph traversal (~2.1 ms) with transparent SQLite 3 embedded multi-join failover (<1 ms) on socket drop.

• **Stage 9 — DBpedia SPARQL Ingestion:** W3C SPARQL 1.1 harvester extracting biographical dates, birthplaces, and local atomic JSON snapshot caching.

• **Stage 10 — Curatorial NLG Engine:** Speech-tailored natural prose generation avoiding key-value dumps and reframing factual discrepancies constructively.

• **Stage 11 — Neural Edge-TTS Voice:** Microsoft Edge Neural TTS (en-GB-RyanNeural, 24 kHz) with offline pyttsx3 synthesis failover within 420 ms.

• **Stage 12 — Client Presentation Hub:** Streamlit reactive web interface rendering native HTML5 audio playback, live discourse state badges, and multi-turn audit transcripts.

12. Knowledge Graph Query Taxonomy & Detailed Query Catalog

The system operates a total of **fifteen distinct computational queries** spanning four distinct operational layers: (1) *Linked Data Ingestion Queries* (2 SPARQL 1.1 queries); (2) *Graph Traversal Queries* (5 Neo4j Cypher templates); (3) *Embedded Relational Queries* (3 SQLite 3 indexed JOINs); and (4) *Contradiction Verification Queries* (5 formal boolean predicates). Table 10 provides a complete catalog of all queries, their target entities, execution engines, and empirical latencies.

Query ID	Layer	Formal Query Type	Target Entities / Logical Predicates	Execution Engine	Latency
Q-SPARQL-1	Ingest	SPARQL 1.1 SELECT	Caravaggio & Battistello biographical properties (dbo:birthDate, deathDate, dbo:abstract)	DBpedia Endpoint	~350 ms
Q-SPARQL-2	Ingest	SPARQL 1.1 SELECT	Artwork URIs, English labels, and museum holdings (dbo:author, dbo:museum)	DBpedia Endpoint	~420 ms
Q-CYPHER-1	Graph	2-Hop Subgraph MATCH	(Artist)-[:PAINTED]->(Artwork)-[:LOCATED_AT]->(Location) full catalog join	Neo4j Bolt Driver	2.10 ms
Q-CYPHER-2	Graph	Filtered MATCH	(Artist {name: $artist})-[:PAINTED]->(Artwork) sub-oeuvre extraction	Neo4j Bolt Driver	1.82 ms
Q-CYPHER-3	Graph	Filtered MATCH	(Artwork)-[:LOCATED_AT]->(Location {name: $location}) venue holding extraction	Neo4j Bolt Driver	1.74 ms
Q-CYPHER-4	Graph	Single Node MATCH	(Artwork {title: $title}) exact and substring property lookup	Neo4j Bolt Driver	1.48 ms
Q-CYPHER-5	Graph	DDL Constraint	CREATE CONSTRAINT FOR (a:Artwork) REQUIRE a.title IS UNIQUE	Neo4j Schema Engine	0.35 ms
Q-SQL-1	Local	2-Hop Table JOIN	SELECT FROM nodes a JOIN edges e1 (PAINTED) JOIN edges e2 (LOCATED_AT)	SQLite 3 Engine	0.92 ms
Q-SQL-2	Local	Indexed Aggregate	SELECT properties FROM nodes WHERE label = 'Location' + artwork count aggregations	SQLite 3 Engine	0.64 ms
Q-SQL-3	Local	Indexed Aggregate	SELECT properties FROM nodes WHERE label = 'Artist' + oeuvre list aggregations	SQLite 3 Engine	0.58 ms
Q-PRED-1	Guard	Attribution Predicate	I_artist(u): Verify claimed artist matches true graph artist attribution	Inconsistency Engine	<0.05 ms
Q-PRED-2	Guard	Location Predicate	I_venue(u): Verify claimed Neapolitan museum matches true graph location	Inconsistency Engine	<0.05 ms
Q-PRED-3	Guard	Spatial Predicate	I_city(u): Intercept external non-Neapolitan masterpieces (Rome, Florence, Paris)	Inconsistency Engine	<0.05 ms
Q-PRED-4	Guard	Scope Predicate	I_scope(u): Detect out-of-scope Italian masters (Bernini, Raphael, Michelangelo)	Inconsistency Engine	<0.05 ms
Q-PRED-5	Guard	Discourse Predicate	I_context(u): Detect pronoun contradiction against prior turn state S(t-1)	Inconsistency Engine	<0.05 ms

12.3 Deterministic Query Selection Tree & Dynamic Routing Logic

The selection and formulation of database queries is governed by a deterministic, multi-stage decision sequence designed to minimize graph overhead while ensuring absolute factual integrity. The decision tree proceeds through five discrete operational phases:

- **Step 1: Contradiction Pre-Check Filter:** The utterance is evaluated against the five inconsistency predicates: I(u) = I_artist v I_venue v I_city v I_scope v I_context. If I(u) = True, execution is immediately diverted away from the knowledge graph: no database queries are dispatched. Instead, the dialogue manager constructs a pedagogical refutation articulating the factual discrepancy and suggesting valid local masterworks.
- **Step 2: Anaphora & Coreference Binding:** If the utterance contains anaphoric pronouns ('it', 'this work', 'where is that') and no explicit artwork entity is detected, the query router checks dialogue state S(t-1). If S_art(t-1) is non-empty and within the recency threshold (W <= 3 turns), the router binds the antecedent artwork entity e_art = S_art(t-1), injecting it into subsequent query generation.
- **Step 3: Entity Hierarchy Dispatch:** When no contradiction is present, the parser dispatches queries based on entity specificity: (a) Artwork Detected -> binds e_art and triggers Q-CYPHER-4 or Q-SQL-1; (b) Location Detected -> binds e_loc and triggers Q-CYPHER-3 or Q-SQL-2; (c) Artist Detected -> binds e_artist and triggers Q-CYPHER-2 or Q-SQL-3; (d) Overview Keywords -> triggers full catalog retrieval; (e) Unmatched -> triggers fallback suggestions.
- **Step 4: Conversational Intent Sub-Routing:** For an identified artwork e_art, the router inspects token intent: (i) Spatial ('where', 'visit') -> selects location & address properties; (ii) Attribution ('who painted') -> selects artist node; (iii) Physical ('dimensions', 'medium') -> selects measurement properties; (iv) Iconography ('meaning', 'significance') -> selects descriptive commentary & DBpedia abstract; (v) Chronology ('year', 'when') -> selects execution date.
- **Step 5: Backend Failover Dispatch:** NeapolitanArtGraph pings the Neo4j Bolt socket. If alive, the parameterized Cypher query (e.g. MATCH (a:Artwork {title: $t})) is executed. If the connection fails or times out (>50 ms), the circuit breaker automatically delegates execution to the embedded SQLite engine (Q-SQL-1), executing the equivalent indexed SQL join with zero perceptible latency change.

12.4 Parameterized Query Execution Templates

Below are the primary parameterized query templates deployed across the primary Neo4j Cypher engine and embedded SQLite 3 failover store:

```
// Cypher Query Template (Parameterized Subgraph Traversal)
MATCH (p:Artist)-[:PAINTED]->(a:Artwork {title: $artwork_title})-[:LOCATED_AT]->(l:Location)
RETURN a.title, a.year, a.medium, a.dimensions, a.description, p.name AS artist, l.name AS location, l.address;

-- SQLite 3 Query Template (Embedded Relational 2-Hop Join Failover)
SELECT a.properties AS art_prop, p.properties AS artist_prop, l.properties AS loc_prop
FROM nodes a JOIN edges e1 ON e1.target_id = a.id AND e1.relation = 'PAINTED'
JOIN nodes p ON p.id = e1.source_id JOIN edges e2 ON e2.source_id = a.id AND e2.relation = 'LOCATED_AT'
JOIN nodes l ON l.id = e2.target_id WHERE a.label = 'Artwork' AND json_extract(a.properties, '$.title') = ?;
```

13. System Deployment, Failure Modes & Engineering Mitigations

The system is architected for zero-configuration reproducibility across server containers and offline edge devices. Standard execution is initialized via shell commands: python3 -m venv .venv && pip install -r requirements.txt, followed by docker run -d -p 7687:7687 neo4j:5.26, python scripts/build_graph.py, and streamlit run app.py --server.port 8502.

Engineering Mitigations: (1) *Acoustic Reverberation:* In high-ceilinged stone churches (>2.5s reverberation), Whisper Small conditioned on token priors maintains high word-error-rate resilience; (2) *Zero-Downtime Failover:* SQLite fallback shields visitors from Neo4j network drops; (3) *Cache Immutability:* SPARQL results are cached atomically to eliminate partial-write corruptions; and (4) *Title Polysemy:* Shortened titles ('Flagellation', 'Ursula') are resolved by intersecting active discourse state with location cues.

14. Conclusion & Strategic Research Trajectories

This monograph establishes that domain-specific conversational systems achieve deterministic factual fidelity when grounded in semantic web knowledge graphs. By integrating live DBpedia SPARQL ingestion, dual graph persistence (Neo4j and SQLite), real-time contradiction reasoning, phonetically conditioned speech recognition, and soft curatorial natural language generation, the system delivers an authoritative, scholarly, and natural conversational guide for the cultural heritage of Naples.

Strategic engineering trajectories include: (1) quantizing Whisper small to INT8/4-bit GGUF models for local NPU execution on low-power museum kiosks; (2) integrating Vision-Language Models (VLM) for real-time camera-based canvas inspection; and (3) expanding the ontological schema to incorporate subsequent Neapolitan Baroque masters including Jusepe de Ribera, Artemisia Gentileschi, Massimo Stanzione, and Luca Giordano.